

Conjecture 00000002491 is false:
in the partition lattice Π_4 , $\mu(\hat{0}, \{12 \mid 34\}) = 1 \neq 0$

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1 The conjecture

Conjecture 00000002491 (English text, verbatim): “Definition: Möbius inversion of partitions: poset inversion. Conjecture: The Möbius function of the partition lattice is governed by squarefree partitions: the inversion formula closes on partitions without repeated parts, and the Möbius values vanish otherwise.” The Chinese text is garbled; it names the same objects (Moebius inversion, the partition lattice, squarefree partitions) and adds no further condition.

Reading. The partition lattice Π_n is the set of all partitions of an n -element set into nonempty blocks, ordered by refinement. A partition σ is “squarefree” (“without repeated parts”) when its type, the integer partition formed by its block sizes, has no repeated part, i.e. no two distinct blocks of σ have the same size. “The Möbius values” are the values $\mu(\hat{0}, \sigma)$ of the Möbius function of Π_n , where $\hat{0}$ is the partition into singletons. The clause refuted is

$$(V) \quad \text{for every } n \text{ and every } \sigma \in \Pi_n \text{ with a repeated block size, } \mu(\hat{0}, \sigma) = 0.$$

We also refute the dual reading (V') in which the values are $\mu(\sigma, \hat{1})$, with $\hat{1}$ the one-block partition. One counterexample refutes these universal clauses, and then the conjunction is false. The vague clause “the inversion formula closes on partitions without repeated parts” is not formalized and not used.

2 Definitions

Definition 1. For a locally finite poset P , the Möbius function is defined recursively by $\mu(a, a) = 1$, $\mu(a, b) = -\sum_{a \leq x < b} \mu(a, x)$ for $a < b$, and $\mu(a, b) = 0$ if $a \not\leq b$.

In Lean this is Mathlib’s `IncidenceAlgebra.mu`; the recursion is Mathlib’s lemma `mu_eq_neg_sum_Ico_of_ne`. The poset Π_4 is Mathlib’s `Finpartition` (`univ : Finset (Fin 4)`) with its refinement order ($P \leq Q$ iff every block of P lies in a block of Q), its bottom $\hat{0}$ (singletons) and its top $\hat{1}$ (one block). We write the ground set as $\{0, 1, 2, 3\}$, so $\sigma = \{01 \mid 23\}$ is the partition $\{12 \mid 34\}$ of the title.

3 Proof

Let $\sigma = \{01 \mid 23\}$, $p_1 = \{01 \mid 2 \mid 3\}$, $p_2 = \{0 \mid 1 \mid 23\}$. The two blocks of σ both have size 2, so σ has a repeated block size.

Lemma 2. *The interval $[\hat{0}, \sigma]$ of Π_4 is $\{\hat{0}, p_1, p_2, \sigma\}$, and the four elements are distinct. Moreover $[\hat{0}, p_1) = [\hat{0}, p_2) = \{\hat{0}\}$.*

Proof. For $x \in \Pi_4$ and a point a , let $x(a)$ be the block of x containing a . Then $x = \{x(0), x(1), x(2), x(3)\}$ as a set of blocks, and $x \leq y$ iff $x(a) \subseteq y(a)$ for every a . If $x \leq \sigma$ then $x(0), x(1) \subseteq \{0, 1\}$. If $1 \in x(0)$ then $x(1) = x(0) = \{0, 1\}$; otherwise $x(0) = \{0\}$, $0 \notin x(1)$ and $x(1) = \{1\}$. The same holds for the points 2, 3 and the block $\{2, 3\}$. The four combinations give $\hat{0}, p_1, p_2, \sigma$. Each of these is $\leq \sigma$, and $\sigma \not\leq p_1$, $\sigma \not\leq p_2$, $p_1 \not\leq p_2$, $p_2 \not\leq p_1$, $p_1 \not\leq \hat{0}$, $p_2 \not\leq \hat{0}$, which shows distinctness and the last claim (an element below p_1 is below σ , hence one of the four, and only $\hat{0}$ is strictly below p_1). $\square$

Theorem 3. $\mu(\hat{0}, \sigma) = 1$ in Π_4 . Hence (V) is false.

Proof. By the recursion and Lemma 2, $\mu(\hat{0}, p_1) = -\mu(\hat{0}, \hat{0}) = -1$, likewise $\mu(\hat{0}, p_2) = -1$, and $\mu(\hat{0}, \sigma) = -(\mu(\hat{0}, \hat{0}) + \mu(\hat{0}, p_1) + \mu(\hat{0}, p_2)) = -(1 - 1 - 1) = 1 \neq 0$. $\square$

Theorem 4. $\mu(\sigma, \hat{1}) = -1$ in Π_4 . Hence (V') is false.

Proof. If $\sigma \leq x$, then $\{0, 1\} \subseteq x(0)$ and $\{2, 3\} \subseteq x(2)$. If $2 \in x(0)$, every point lies in $x(0)$ and $x = \hat{1}$. Otherwise $3 \notin x(0)$ (else $x(0) = x(3) = x(2) \ni 2$), and similarly $0, 1 \notin x(2)$, so $x(0) = x(1) = \{0, 1\}$, $x(2) = x(3) = \{2, 3\}$ and $x = \sigma$. So $[\sigma, \hat{1}) = \{\sigma\}$ ($\hat{1} \not\leq \sigma$) and $\mu(\sigma, \hat{1}) = -\mu(\sigma, \sigma) = -1$. $\square$

Consistency check (not used in Lean). The retrieved source below gives, for $\sigma \leq \tau$ in the partition lattice with the t blocks of τ splitting into $s_1, \dots, s_t$ blocks of σ and $s = s_1 + \dots + s_t$, $\mu(\sigma, \tau) = (-1)^{s-t} (s_1 - 1)! \dots (s_t - 1)!$. For $(\hat{0}, \sigma)$ this gives $(-1)^2 1! 1! = 1$, and for $(\sigma, \hat{1})$ it gives $(-1)^1 1! = -1$, matching Theorems 3 and 4. The formula also shows that $\mu(\sigma, \tau)$ is never 0 when $\sigma \leq \tau$, so the vanishing clause fails at every repeated-type partition, but our proof uses only the single explicit counterexample.

4 Lean formalization

File `lean/Conjecture2491/Basic.lean` (317 lines, only import `Mathlib`), namespace `C2491`. Objects: `Pi4 := Finpartition (univ : Finset (Fin 4))` (Mathlib's partitions with the refinement order), `IncidenceAlgebra.mu` over $\mathbb{Z}$ (Mathlib's Möbius function), the partitions `sigma`, `p1`, `p2` given by their blocks, and

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HasRepeatedBlockSize P := exists B in P.parts, exists C in P.parts, B != C and #B
                        = #C.
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The `LocallyFiniteOrder` instance is `Fintype.toLocallyFiniteOrder`; this choice does not matter, since `LocallyFiniteOrder` is a subsingleton, and the proofs use only membership in intervals. Main theorems:

- `mu_bot_sigma : mu Z (bot : Pi4) sigma = 1` (Theorem 3).
- `conjecture_2491_false : not (forall s : Pi4, HasRepeatedBlockSize s -> mu Z bot s = 0)`, i.e. (V) fails.
- `mu_sigma_top : mu Z sigma (top : Pi4) = -1` and `conjecture_2491_false_dual` (Theorem 4, (V') fails).

Supporting lemmas: `parts_eq` and `le_iff` (blocks through points), `pair_cases`, `le_sigma_cases` and `Ico_bot_sigma`, `Ico_bot_p1`, `Ico_bot_p2` (Lemma 2), `sigma_le_cases` and `Ico_sigma_top` (Theorem 4). The order facts between the explicit partitions are decided by `decide` on their explicit block lists. The interval enumeration itself is proved by hand for an arbitrary partition; it is not a finite search over Π_4 . `#print axioms` for each main theorem gives `[propext, Classical.choice, Quot.sound]`; there is no `sorry` and no `native_decide`. Build: `lake build` with Mathlib v4.33.1.

5 Relation to the earlier submission

An earlier submission (PR #191, orionsheep) made the same mathematical point but was closed. The reviewer wrote: “muAux runs on a hardcoded 4×4 Boolean matrix with no partitions and no partition lattice defined in Lean; the identification with the interval $[\hat{0}, \sigma]$ exists only in comments, and the padding theorem ($2 = 2 : \text{Bool}$) confirms no conjecture object is formalized.” This submission formalizes the actual objects: Mathlib’s `Finpartition` of a 4-set with its refinement order, Mathlib’s Möbius function `IncidenceAlgebra.mu`, and the predicate “has a repeated block size”. The enumeration of $[\hat{0}, \sigma]$ is a Lean theorem about all partitions $x \leq \sigma$ (`le_sigma_cases`). The earlier prose also said the $(2, 1, 1)$ type occurs twice in Π_4 ; it occurs six times, and that count is not used here.

Source

Wikipedia, “Incidence algebra”, section “Partitions of a set”, https://en.wikipedia.org/wiki/Incidence_algebra: “Partially order the set of all partitions of a finite set by saying $\sigma \leq \tau$ if σ is a finer partition than τ . In particular, let τ have t blocks which respectively split into $s_1, \dots, s_t$ finer blocks of σ , which has a total of $s = s_1 + \dots + s_t$ blocks. Then the Möbius function is: $\mu(\sigma, \tau) = (-1)^{s-t}(s_1 - 1)! \dots (s_t - 1)!$ ” Used only for the consistency check.